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Partitioning of macroinvertebrate assemblages across temporary pools in an intermittent dryland river

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ABSTRACT

Streams in arid regions are seen as mosaics of spatial/temporal landscapes, with high environmental heterogeneity and exerting strong influence on patterns, structure, and diversity of biotic communities. This study describes the richness and density of benthic macroinvertebrates in temporary pools across different sections of a dryland intermittent river catchment and characterizes the diversity partitioning at different spatial scales across the longitudinal gradient of the river. We hypothesized that the diversity of invertebrates is not homogeneously distributed among the different spatial scales but concentrated at the finer levels of pools and river reaches. The study was conducted in an intermittent river catchment in semiarid Brazil. Sampling was performed hierarchically, including 3 segments of the river (upper, middle, and a tributary), 3 reaches per segment, 3 sites per reach, and 3 microhabitats per site ($n = 81$). Macroinvertebrate diversity was evaluated at scales of river segment (β_4), reach (β_3), site (β_2), and microhabitat (β_1). Habitat structure was composed of a diverse array of elements that varied spatially. Scale-dependent spatial segregation of community composition was observed on macroinvertebrate assemblages. Diversity partitioning analysis reinforced the importance of broader spatial scales (β_4) and lesser importance of finer spatial scales (β_1 and β_2) for the regional diversity in the dryland river catchment study site. Our observations reflect a combination of homogenization of assemblage composition and reduced diversity at finer scales, with a greater contribution of broader scales to the diversity.

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Introduction

Dryland rivers are often characterized by intermittent surface water flow, resulting in the expansion and contraction of the river network and a spatial/temporal mosaic of lotic, lentic, and terrestrial habitats (Williams 1999, Steward et al. 2011, Datry et al. 2014, Godsey and Kirchner 2014). These hydrological dynamics have been reported to structure the physical and chemical environmental characteristics (Marshall et al. 2006) and to segregate the diversity and distribution of aquatic organisms into isolated patches of habitat (e.g., pools, riffles, macrophyte stands; Sheldon et al. 2002, Humphries and Baldwin 2003, Marshall et al. 2006, Mugodo et al. 2006). These small and patchy systems are expected to contribute greatly to the regional pool of species diversity given the high variability among local assemblages (De Meester et al. 2005). Species would be filtered from the regional pool in accordance with their biological traits (such as habitat use and dispersal patterns and ability) and specific adaptations and strategies to cope with the variable and commonly harsh environmental conditions (Quist et al. 2005).

These processes create a dynamic and complex array of interactions between the intermittent river fauna and the environmental conditions regulated by factors at local (within pools or river reaches) and regional (catchment) scales (Medeiros et al. 2008, Larned et al. 2010). Organisms in such heterogeneous systems are thus expected to be distributed in patches created by the environmental conditions, which in turn affect the fauna hierarchically (Stewart-Koster et al. 2007).

Freshwater macroinvertebrates have strong interaction with the environmental conditions in rivers and streams, and their patterns of distribution can be traced to the types and dynamics of habitat available (Li et al. 2012, Carvalho et al. 2013). As a consequence, the diversity is filtered in a hierarchical fashion, from the distinct river morphological zones (macrohabitats; Thoms et al. 2004), through the pools, runs, and riffles (mesohabitats), and to the stands of macrophytes, submerged vegetation, and substrate types (microhabitats; Frissell et al. 1986, Pringle et al. 1988, Leibold et al. 2004).

Therefore, because the diversity of species and community composition within a given scale can be different from those observed at a different scale, the diversity is often split or partitioned into its constituent components to compare and describe its relative contribution across each scale and to better understand the organization of the biological community (Gotelli and Colwell 2001, Crist et al. 2003). In general, the diversity of species within local or single homogeneous spots is termed alpha (α) diversity. At larger spatial scales, regional diversity is described as gamma (γ) diversity. A third component, beta (β) diversity, emerges when diversity is compared within and/or between the different scales (Whittaker et al. 2001).

A range of studies has been using the method of additive partitioning to comprehend the contribution of each diversity component to the spatial organization of freshwater macroinvertebrates in temperate perennial river systems (Stendera and Johnson 2005, Ligeiro et al. 2010, Ávila et al. 2011, Shell and Collier 2018). Dryland intermittent rivers and streams constitute >30% of the total length and discharge of the global river network (Tooth 2000, Detry et al. 2014); therefore, knowledge of the partitioning of diversity in these systems will aid in the understanding of how communities are structured in highly variable and spatially segregated ecosystems. In semiarid Brazil, these communities are biologically diverse, and rivers and streams are dynamic systems representing distinctive riverscape features within the dry shrubby deciduous open forest they flow through; yet their natural flow regime and ecological and morphological integrity are threatened by the expansion of water resource developments and interbasin water transfers (Maltchik and Medeiros 2006).

This study describes the richness and density of macroinvertebrates in temporary pools across different sections of a dryland intermittent river and uses hierarchical additive partitioning to evaluate the relative contribution of different spatial scales to the diversity of macroinvertebrate assemblages. Given the expected patchy nature of intermittent rivers (Melo and Medeiros 2013), we hypothesized that the diversity is not homogeneously distributed among the different spatial scales but concentrated at the finer levels of pools and river reaches. Some environmental features (reach morphology, habitat structure, and water quality) are also described to explore possible explaining factors for the observed patterns.

Materials and methods

Study area and sampling design

This study was performed at the Paraíba River catchment, a dryland intermittent river in the northeast of

Brazil. The main hydrological feature in Brazilian semiarid rivers is the intermittence of surface water flow, which leads to a network of dry riverbeds scattered with temporary and semipermanent pools (Maltchik and Medeiros 2006). This pattern is the result of low average precipitation, which is irregular and concentrated in the warmer months, usually between January and May (Araújo et al. 2008), and high average annual temperatures between 20 and 32 °C. These rivers flow through a dry and shrubby deciduous open forest, characterized by the presence of xerophytic species (Leal et al. 2005). The climate in the study area is semiarid (*BS'h*, hot and dry) and equatorial (*As*, dry summer; Peel et al. 2007 updated) with average temperatures of 25 °C and annual precipitation ~600 mm.

Sampling was performed across 3 segments of the Paraíba River catchment: 2 upper segments, represented by the Taperoá River (an affluent of the Paraíba River) and the upper Paraíba River (west of the Epitácio Pessoa reservoir), and a middle segment east of the reservoir (Fig. 1). The lower reaches were not sampled to avoid marine coastal influence on the freshwater macroinvertebrate taxa composition. Sampling was performed in March 2015 during the rainy season. Within each of the 3 river segments, 3 reaches were sampled ($n = 9$), within each reach 3 sites (pool/riffle system) were sampled ($n = 27$), and within each site 3 microhabitats were sampled ($n = 81$). Microhabitats represented different substrate types, usually sand, mud, and macrophyte stands. Segment, reach, pool/riffle, and microhabitat systems are based on the hierarchical framework for stream habitats proposed by Frissell et al (1986).

At each sampling site, supportive environmental data were measured, comprising (1) physical and chemical water variables, (2) reach morphometry, (3) substrate composition, and (4) the physical structure of the habitat. The physical and chemical variables were estimated using a portable multiparameter water quality meter (Hanna HI 9828, Smithfield, RI, USA) for pH, conductivity ($\mu\text{S}/\text{cm}$), dissolved oxygen (mg/L), and temperature (°C). Transparency (cm) was measured with a Secchi disk, and water velocity (m/s) was measured using the float method (Maitland 1990). River reach morphology was evaluated as the reach altitude (m) and the average width (m) and depth (cm) taken from 3 random transects. The substrate composition and habitat structure were estimated in 12 survey points, each 1 m^2 , established randomly along the margins at the aquatic–terrestrial interface (Medeiros et al. 2008). At each survey point, the proportion of the sediment composition (mud, sand, gravel, and cobbles) and littoral and underwater structures (e.g. macrophytes, littoral grass, submerged vegetation, overhanging vegetation, leaf litter, algae, and woody debris) were estimated visually.

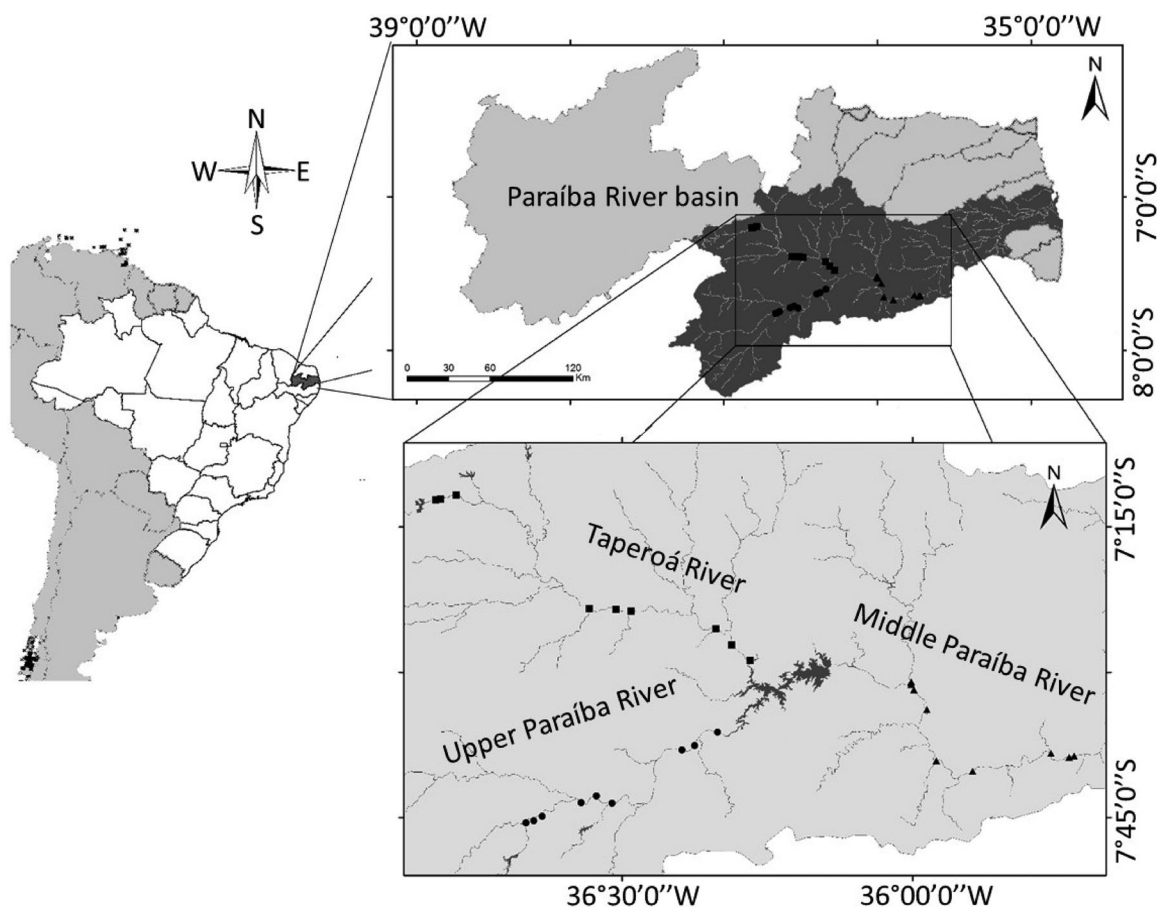


Figure 1. Study area showing the Paraíba River basin and the sites sampled during the present study.

Macroinvertebrate sampling and statistical analyses

At each sampling site, 3 samples of sediment (40 cm × 40 cm area and 2.5 cm depth) were taken with a D-shaped net (40 cm wide, 250 µm mesh) to represent the different microhabitat types (the finest study scale). Samples were fixed in 4% formalin in the field and taken to the laboratory where they were wet sieved (250 µm) and preserved in 70% alcohol. Macroinvertebrate individuals were sorted, and all organisms other than Chironomidae were identified to the lowest possible taxonomic level (usually family; McCafferty 1983, Williams and Feltmate 1992, Mugnai et al. 2010) and counted. A subsample of Chironomidae larvae was counted (Baker and Huggins 2005, Farias et al. 2015a), and individuals were identified to the level of genus after being mounted onto semipermanent slides using Hoyer's medium (Trivinho-Strixino 2011). Chironomidae larvae are the most abundant and diverse group of insects in intermittent streams, and their densities are much higher than those of macroinvertebrates. Therefore, they require more precise identification and separate statistical analysis to prevent bias in the analysis of

the macroinvertebrate community (Farias et al. 2012, Rocha et al. 2012). In the present study a separate analysis was performed on Chironomidae genera to evaluate whether their patterns of diversity partitioning are similar to the macroinvertebrate fauna as a whole.

The evaluation of the community structure as well as all statistical analyses were performed on density (individuals per square meter), calculated as the number of individuals divided by the sampled area of the D-shaped net. Richness and density of taxa were used to describe patterns of distribution across assemblages. Life stages (larvae, pupae, and adult) were treated as separate taxa given their differences in resource and habitat use (Carvalho et al. 2013). Significance of differences for richness and density across spatial scales were measured using analysis of variance (ANOVA) followed by Tukey's post hoc test for pairwise comparisons ($\alpha = 0.05$; Zar 1999, Maltchik et al. 2010).

A hierarchical flexible beta clustering ($\beta = -0.25$) based on Bray-Curtis distance was used to assess dissimilarities in assemblage composition across different spatial scales (Ávila et al. 2011). The interpretation of the groupings in the dendrogram was subjective, and the

decision about which groups to report was based on the arbitrary cut-off value of up to 25% of information remaining (Quinn and Keough 2002). Density data were relativized and square-root transformed prior to statistical analyses to enhance normality and equality of variances of the dataset (Rocha et al. 2012). The multi-response permutation procedure (MRPP) was used to test for significance of differences in taxa composition across spatial scales (Biondini et al. 1985, McCune and Grace 2002). The value of A is presented as a measure of the degree of homogeneity between groups compared to random expectation. Clustering and MRPP were performed on PC-ORD 7.07 (McCune and Mefford 2018).

Additive partitioning was used to decompose the total variation in community composition (river catchment diversity) into its α and β components, where α diversity represents the within-unit diversity and β diversity is the between-unit diversity within a given scale (Lande 1996, Crist et al. 2003). Macroinvertebrate and Chironomidae diversity were analyzed separately at the 4 different spatial scales, and 1 value of β diversity was associated with each spatial level: at the fine microhabitat, spatial scale was the individual samples at each substrate type ($n = 81$); at the intermediate, spatial scales were the 3 pooled substrate types collected at each site or pool/riffle ($n = 27$) and the 3 reaches with pooled sites ($n = 9$); and at the broad spatial scale were the 3 segments of the river catchment ($n = 3$). Therefore, the total diversity (γ) was partitioned into the diversity (expressed as a percentage) within microhabitats (α_1 , defined as the average richness per substrate type) and between microhabitats (β_1 , represented by the differences of taxa collected across substrate types), between pool/riffles (β_2), between river reaches (β_3), and between river segments (β_4). Therefore, the total diversity is expressed as the proportional contributions of diversity at each level in the hierarchical sampling design, with the total (γ) diversity represented by $\alpha_1 + \beta_1 + \beta_2 + \beta_3 + \beta_4$ (Lande 1996, Crist et al. 2003, Melo and Medeiros 2013).

Diversity partitioning analyses were performed using the additive partitioning function *adipart* in the *vegan* package in R (Oksanen et al. 2017, R Core Team 2013), with individually based randomizations (type I null model; Crist et al. 2003). The randomization process was repeated 10 000 times to obtain null distributions of the α and β diversity estimates at each hierarchical level. The probabilities obtained from randomization tests were interpreted as p values in the sense that a low value ($p < 0.05$) indicates that the observed diversity is significantly higher than that expected under the null model, and a high value ($p > 0.95$) indicates that the observed diversity is significantly lower than that

expected (Gering et al. 2003, Veech et al. 2003, Veech and Crist 2009).

Results

Environmental variables

Rainfall during the present study was highly variable, with more regular values of precipitation recorded throughout June and July (Fig. 2). As a consequence, surface water flow was observed intermittently during the study period and across the river catchment, with only some reaches registering water velocity (Table 1). River widths were consistent with a constrained river, like the one in our study, which does not overflow into a floodplain, and depths showed relatively shallow habitats with values ≤ 56 cm. In general, waters were neutral to alkaline (pH 7.6–9.4) and well oxygenated (5.2–7.9 mg/L). Water temperatures ranged from 27.8 to 29.3 °C. Transparency was low across all reaches, with Secchi depths between 19.9 and 37.9 cm, and conductivity was high, up to 4919.5 ($\mu\text{S}/\text{cm}$) at the lower river reaches (Table 1). The structure of the littoral habitat was diverse, composed mostly of woody debris, submerged vegetation, littoral grass, and aquatic macrophytes. Woody debris was abundant at all segments of the river catchment, whereas submerged vegetation was abundant at the higher reaches. The substrate was mainly composed of mud (42.2–66.8%) and sand (21.4–53.8%; Table 1).

Macroinvertebrate and Chironomidae assemblages and diversity

We collected 82 782 macroinvertebrate individuals during the present study, distributed across 33 taxa and life stages. Insecta was the most important class, with 20 families. Among the Insecta, Diptera and Gastropoda were each represented by 6 families and Hemiptera by 5 families. The Chironomidae family was the most abundant with 58.8% of the individuals collected, followed by Thiaridae (Mollusca, Gastropoda) with 25.7% of all individuals (Fig. 3). Even though some taxa were present in all 3 study segments, several families of macroinvertebrates were specific to some segments of the study catchment: Naucoridae, Veliidae, Physidae, and Sphaeriidae were recorded only at the upper segment; Culicidae, Stratiomyidae, Belostomatidae, Notonectidae, and Coenagrionidae were recorded only at the Taperoá segment; and Elmidae, Chaoboridae, Odontoceridae, Hydracarina, Atyidae, and Corbiculidae were recorded only at the middle segment of the catchment (Table 2).

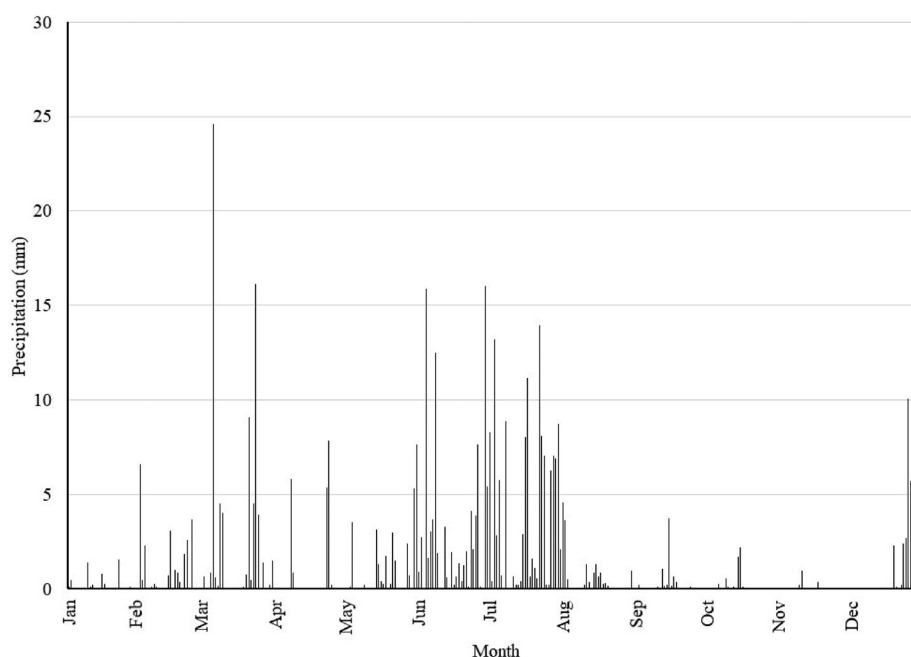


Figure 2. Daily precipitation in the study area (7°23'27"S, 36°31'58"W; São João do Cariri Meteorological Station) from January to December 2015. Source: <http://www.cptec.inpe.br/proclima>.

Table 1. Average proportional cover of habitat elements, morphometry, and physical and chemical variables (standard deviation) in the Paraíba River during the 2015 wet season.

	Upper segment	Taperoá segment	Middle segment	Min-max
Habitat structure (%)				
Macrophyte	5.65 (16.18)	0.05 (0.14)	3.05 (5.24)	3.05–16.20
Littoral grass	0.42 (0.72)	6.27 (12.23)	12.41 (13.75)	0.42–12.41
Submerged vegetation	16.20 (10.67)	7.80 (12.01)	0.56 (1.23)	0.56–16.20
Overhanging vegetation	2.82 (5.52)	1.85 (2.60)	0	0.00–2.82
Leaf litter	2.36 (3.12)	1.52 (2.32)	1.20 (1.48)	1.20–2.36
Algae	0.37 (0.70)	5.51 (7.29)	1.90 (3.78)	0.37–5.51
Woody debris	8.10 (5.32)	3.75 (5.46)	8.01 (5.37)	3.75–8.10
Substrate composition (%)				
Mud	42.17 (33.30)	45.97 (45.56)	66.80 (32.19)	42.17–66.80
Sand	53.84 (31.93)	48.29 (41.87)	21.43 (25.37)	21.43–53.84
Gravel	1.62 (1.45)	1.34 (1.35)	3.61 (3.28)	1.62–3.61
Cobble	2.36 (3.94)	0.60 (1.30)	1.34 (1.26)	0.60–2.36
Rocks	0	1.62 (3.57)	6.80 (8.70)	0.00–6.80
Boulders	0	2.13 (6.39)	0	0.00–2.13
Physical and chemical water variables				
Temperature (°C)	27.84 (2.59)	28.86 (2.57)	29.30 (2.13)	27.84–29.30
Conductivity (µS/cm)	315.70 (159.71)	650.15 (252.42)	4919.48 (3465.65)	315.70–4919.48
Dissolved oxygen (mg/L)	5.20 (1.73)	6.78 (1.53)	7.95 (1.16)	5.20–7.95
Transparency (cm)	29.88 (20.45)	37.96 (12.22)	19.90 (6.41)	19.90–37.96
pH	7.66 (0.42)	8.35 (0.52)	9.41 (0.38)	7.66–9.41
Reach morphometry				
Altitude (m)	443.21 (37.17)	455.01 (51.71)	252.47 (73.46)	252.47–455.01
Depth (cm)	42.51 (17.42)	55.88 (30.20)	52.91 (29.45)	42.51–55.88
Water velocity (m/s)	0.018 (0.026)	0.045 (0.094)	0	0.00–0.261
Width (m)	20.26 (16.23)	22.02 (24.41)	35.99 (44.75)	20.26–35.99

Min-max = minimum-maximum.

Among the Chironomidae, a subsample of 7515 individuals was analyzed, represented by 3 subfamilies (Chironominae, Tanypodinae, and Orthocladiinae) and 25 genera. *Paratanytarsus* (31.3%) and *Chironomus* (26.6%) were the most abundant genera and were also

present in all study river segments (Fig. 3). The least common and abundant genera were *Alotanypus*, *Corynoneura*, *Fissimentum*, and *Paratendipes*, which were recorded only at the upper segment; *Pelomus* and *Cricotopus*, which occurred only at the Taperoá segment; and

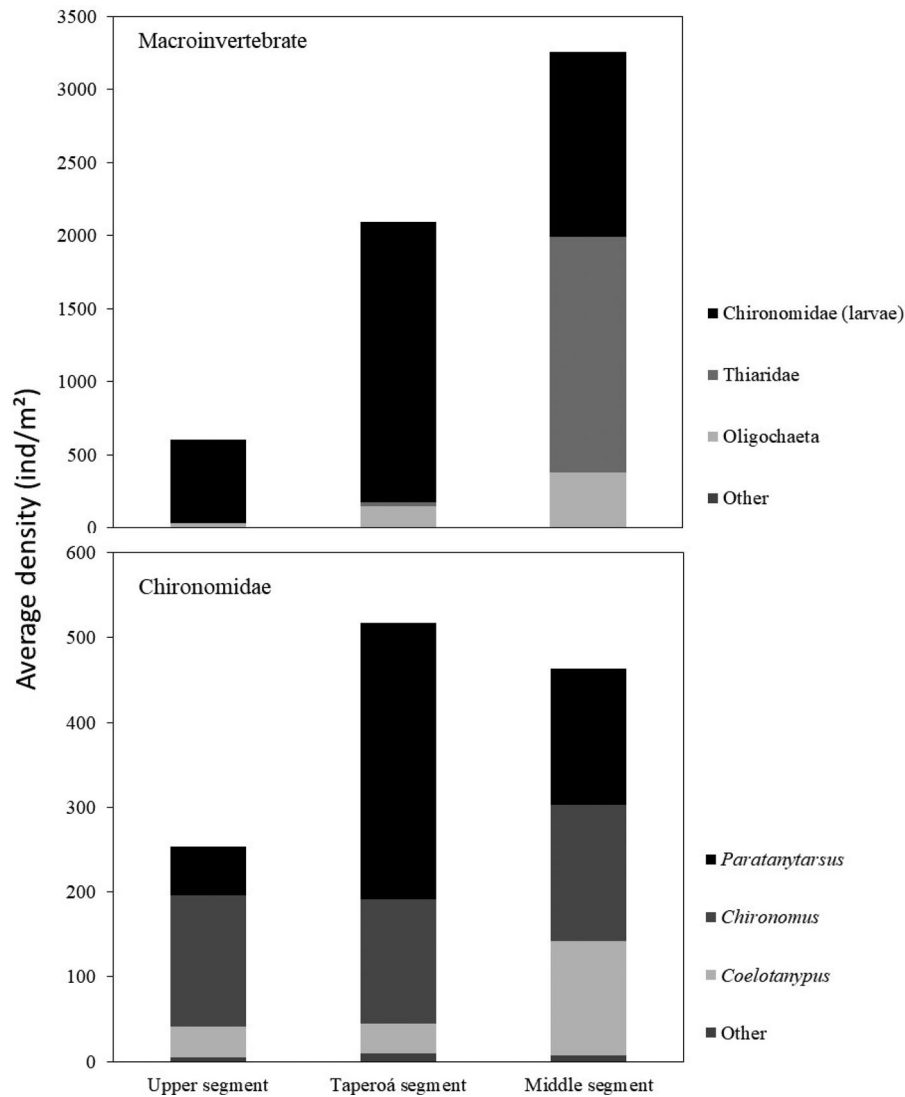


Figure 3. Average density of macroinvertebrates in the Paraíba River during the 2015 wet season.

Tanytus, which occurred only at the middle segment (Table 3).

Even though the macroinvertebrate richness was not significantly different among the 3 study segments (ANOVA $F_{2,24} = 0.889$, $p = 0.424$), the density of organisms was significantly different (ANOVA $F_{2,24} = 10.135$, $p = 0.001$), where the upper segment had lower densities than the Taperoá (Tukey's post hoc $p = 0.010$) and the middle segments (Tukey's post hoc $p = 0.001$). Among the Chironomidae, richness and densities were significantly different across river segments (ANOVA richness $F_{2,23} = 5.265$, $p = 0.013$; ANOVA density $F_{2,23} = 3.413$, $p = 0.050$), where the Taperoá segment showed greater richness than the middle segment (Tukey's post hoc $p = 0.013$), and the upper segment showed lower densities than the Taperoá segment (Tukey's post hoc $p = 0.042$).

Cluster analysis for macroinvertebrates showed 3 groups between 0% and 25% of information remaining

(Fig. 4), representing the broader spatial scales of river segments and reaches. Within segments, the middle reaches tended to segregate from the upper and Taperoá reaches, leading to significant differences in assemblage composition among river segments (MRPP $A = 0.054$, $p = 0.00009$) and among reaches (MRPP $A = 0.030$, $p = 0.008$). These differences were mostly between reaches across the upper and middle segments (pairwise MRPP $A = 0.058$, $p = 0.00008$) and the Taperoá and middle segments (pairwise MRPP $A = 0.058$, $p = 0.0006$). Cluster analysis for Chironomidae yielded results similar to those of macroinvertebrates, where groups between 0% and 25% of information remaining (Fig. 4) represented the broader spatial scales, segregating Chironomidae assemblage composition mainly across river segments (MRPP $A = 0.057$, $p = 0.002$) and reaches (MRPP $A = 0.046$, $p = 0.007$). Similar to the results for macroinvertebrates, pairwise comparisons showed differences

Table 2. Average density (ind/m²) (standard deviation) and frequency of occurrence (F.O., %) of macroinvertebrates in the Paraíba River during the 2015 wet season.

Taxa	Upper segment	Taperoá segment	Middle segment	F.O. (n = 27)
Coleoptera				
Dytiscidae (larvae)	2.31 (3.44)	6.79 (10.83)	0	48.15
Dytiscidae (adult)	0.15 (0.31)	0	0	7.41
Elmidae (adult)	0	0	0.23 (0.69)	3.70
Hydrophilidae (larvae)	3.40 (6.50)	13.50 (19.74)	14.20 (29.01)	62.96
Diptera				
Ceratopogonidae	36.34 (39.48)	33.33 (32.05)	26.23 (71.98)	96.30
Ceratopogonidae (pupae)	0.46 (0.98)	6.80 (19.60)	0.54 (1.08)	25.93
Culicidae	0	0.08 (0.23)	0	3.70
Chaoboridae	0	0	2.93 (8.80)	3.70
Chironomidae	570.91 (675.47)	1924.80 (1071.50)	1263.19 (1515.54)	100.00
Chironomidae (pupae)	7.56 (8.21)	26.62 (18.40)	27.39 (33.39)	92.59
Stratiomyidae	0	0.08 (0.23)	0	3.70
Tipulidae	1.47 (3.63)	0	5.25 (15.74)	18.52
Ephemeroptera				
Baetidae	22.30 (38.73)	17.59 (28.61)	0.08 (0.23)	59.26
Caenidae	34.10 (72.44)	65.60 (102.75)	7.01 (20.26)	51.85
Hemiptera				
Belostomatidae	0	0.08 (0.23)	0	3.70
Corixidae	17.59 (32.30)	4.48 (7.12)	0.38 (0.70)	51.85
Naucoridae	5.40 (16.20)	0	0	3.70
Notonectidae	0	0.77 (1.70)	0	7.41
Veliidae	0.08 (0.23)	0	0	3.70
Odonata				
Coenagrionidae	0	0.15 (0.31)	0	7.41
Gomphidae	1.08 (1.81)	2.24 (2.64)	0.46 (0.49)	59.26
Libellulidae	4.40 (5.58)	10.03 (10.00)	3.63 (6.90)	70.37
Trichoptera				
Odontoceridae	0	0	0.43 (1.39)	3.70
Acari				
Hidracarina	0	0	0.08 (0.23)	3.70
Crustacea				
Atyidae	0	0	0.08 (0.23)	3.70
Conchostraca	0.62 (1.07)	0.77 (1.45)	0.08 (0.23)	25.93
Gastropoda				
Corbiculidae	0	0	0.15 (0.46)	3.70
Planorbidae	0.23 (0.69)	12.11 (35.56)	13.90 (17.60)	48.15
Physidae	0.07 (0.23)	0	0	3.70
Sphaeriidae	0.15 (0.46)	0	0	3.70
Thiaridae	1.16 (2.0)	22.07 (63.40)	1616.44 (1708.64)	59.26
Nematoda	0.23 (0.49)	1.00 (2.76)	0.15 (0.31)	22.22
Oligochaeta	27.24 (33.75)	144.67 (385.62)	373.77 (681.11)	100.00

between the upper and middle segments (pairwise MRPP $A = 0.054$, $p = 0.005$) and the Taperoá and middle segments (pairwise MRPP $A = 0.057$, $p = 0.0024$) for the Chironomidae assemblages.

Partitioning of α and β components showed similar patterns for both macroinvertebrate and Chironomidae diversities. Regarding the fine spatial scale, additive partitioning showed that across microhabitats (α_1 [within microhabitats] and β_1 [among microhabitats]), the observed diversity was significantly lower than that expected from the null model (macroinvertebrates α_1 $p = 0.001$ and β_1 $p = 0.005$; Chironomidae α_1 $p = 0.001$ and β_1 $p = 0.011$), with average observed and expected α_1 of 7.3 and 13.1 and β_1 of 3.3 and 3.9, respectively, for macroinvertebrates, and average observed and expected α_1 of 5.6 and 11.1 and β_1 of 2.6 and 3.0, respectively, for Chironomidae. The spatial scales represented by the β_1 component explained the smallest proportion of the total macroinvertebrate (10.1%) and

Chironomidae (10.5%) diversity (Fig. 5). At the intermediate spatial scales (β_2 [among pool/riffles] and β_3 [among river reaches]), the observed diversity was greater than that expected from the null model (β_2 $p = 0.001$ and β_3 $p = 0.001$) for macroinvertebrates but equal to that expected from the null model for the Chironomidae β_2 component (β_2 $p = 0.155$) and greater than that expected from the null model for the Chironomidae β_3 component (β_3 $p = 0.001$). For macroinvertebrates, the average observed and expected β_2 were 4.8 and 3.8, respectively, and 6.2 and 4.9, respectively, for β_3 . These spatial scales represented 14.6% and 18.9% the total macroinvertebrate diversity. Regarding the Chironomidae, the average observed and expected β_2 were 3.6 and 3.2, respectively, and 5.4 and 3.7, respectively, for β_3 . These spatial scales represented 14.5% and 21.8% the total Chironomidae diversity. At the broadest spatial scale studied (β_4 [among segments of the river catchment]), the observed diversity of

Table 3. Average density (ind/m²) (standard deviation) and frequency of occurrence (F.O., %) of Chironomidae in the Paraíba River during the 2015 wet season.

	Upper segment	Taperoá segment	Middle segment	F.O. (n = 27)
Chironominae				
<i>Aedokritus</i>	1.39 (5.00)	5.56 (13.58)	12.73 (31.84)	37.04
<i>Asheum</i>	0.23 (1.20)	0.69 (3.61)	0	7.41
<i>Apedilum</i>	6.02 (13.03)	19.21 (38.68)	0	33.33
<i>Chironomus</i>	155.09 (232.39)	146.30 (231.51)	1160.65 (246.26)	85.19
<i>Dicrotendipes</i>	7.64 (14.54)	0.46 (1.67)	10.42 (52.90)	25.93
<i>Fissimentum</i>	0.23 (1.20)	0	0	3.70
<i>Goeldichironomus</i>	8.57 (21.24)	34.49 (56.47)	32.87 (65.06)	70.37
<i>Parachironomus</i>	8.33 (12.74)	11.57 (20.78)	0	51.85
<i>Paratanytarsus</i>	57.18 (76.11)	325.93 (409.41)	160.88 (327.34)	88.89
<i>Paratendipes</i>	0.23 (1.20)	0	0	3.70
<i>Polypedilum</i>	18.06 (38.74)	46.60 (64.92)	18.98 (30.54)	77.78
<i>Pelomus</i>	0	0.46 (1.67)	0	7.41
<i>Saetheria</i>	12.55 (25.77)	46.06 (44.50)	96.76 (152.04)	77.78
<i>Stenochironomus</i>	0.23 (1.20)	3.01 (15.64)	0	7.41
Tanypodinae				
<i>Ablabesmyia</i>	10.88 (26.66)	26.39 (43.69)	0.69 (2.65)	33.33
<i>Alotanypus</i>	1.39 (4.36)	0	0	3.70
<i>Brundiniella</i>	0.46 (2.40)	0.23 (1.20)	0	7.41
<i>Coelotanypus</i>	36.11 (82.70)	35.88 (55.43)	134.26 (205.15)	74.07
<i>Clinotanypus</i>	0.69 (3.61)	2.08 (10.82)	0.92 (3.34)	14.81
<i>Djalmabatista</i>	9.72 (38.30)	0	0	18.52
<i>Labrundinia</i>	0	5.56 (14.01)	0	14.52
<i>Procladius</i>	41.20 (59.82)	21.30 (29.33)	1.39 (3.61)	62.96
<i>Tanypus</i>	0	0	0.69 (2.65)	7.41
Orthoclaadiinae				
<i>Corynoneura</i>	0.46 (2.40)	0	0	3.70
<i>Cricotopus</i>	0	0.23 (1.20)	0	3.70

macroinvertebrates was also greater than that expected from the null model ($\beta_4 p = 0.001$), with average observed and expected β_4 of 11.3 and 7.3, respectively. These spatial scales represented 34.3% of the total macroinvertebrate diversity. Similarly, the observed diversity of Chironomidae at the broadest spatial scale was also greater than that expected from the null model ($\beta_4 p = 0.001$), with average observed and expected β_4 of 7.7 and 3.9, respectively. These spatial scales represented 30.7% of the total Chironomidae diversity.

Discussion

Given the expected patchy nature of the study river system, we hypothesized in the present study that the diversity is not homogeneously distributed among the different spatial scales but concentrated at the finer levels of pools and river reaches. Environmental features associated with these scales (reach morphology, habitat structure, and water quality) need to be explored as possible explanatory factors for any observed patterns.

Although the present study was performed during the rainy season, the Paraíba River catchment presented fragmented surface water flow throughout its basin, with only some reaches showing water discharges, resulting in the slow average water velocities observed. This fragmentation is a natural feature of dryland intermittent rivers (Leal et al. 2005, Medeiros and Arthington 2011)

and, in the present study, was the result of scattered rainfall throughout the study area, leading to a system composed mostly of isolated temporary pools rather than the pool/riffle systems proposed by Frissell et al. (1986) for permanent rivers.

The richness and composition of macroinvertebrates observed in the present study are in accordance with other studies developed in semiarid Brazil (Farias et al. 2012, Rocha et al. 2012, Carvalho et al. 2013, Farias et al. 2015b) and elsewhere (Pires et al. 2000, Acuña et al. 2005), represented mostly by Chironomidae and, in the study area, by Thiaridae. The Chironomidae diversity and composition have been reported to respond to varying discharges, depths, and stream widths as well as dissolved oxygen and conductivity (Acuña et al. 2005, Rocha et al. 2012), more significantly so in highly variable systems such as intermittent rivers, which show not only temporal variation in environmental variables but also spatially segregated habitat structure (Medeiros et al. 2008, Carvalho et al. 2013). These organisms show greater resistance to flooding and pioneer the recolonization after flooding (Maltchik and Florin 2002), which makes them highly adaptable to the environmental conditions observed in intermittent rivers, therefore showing ample distribution. Among the most commonly observed macroinvertebrates, the Thiaridae gastropod was represented by *Melanoides tuberculatus*, an invasive species reported to dominate benthic communities with no predation and

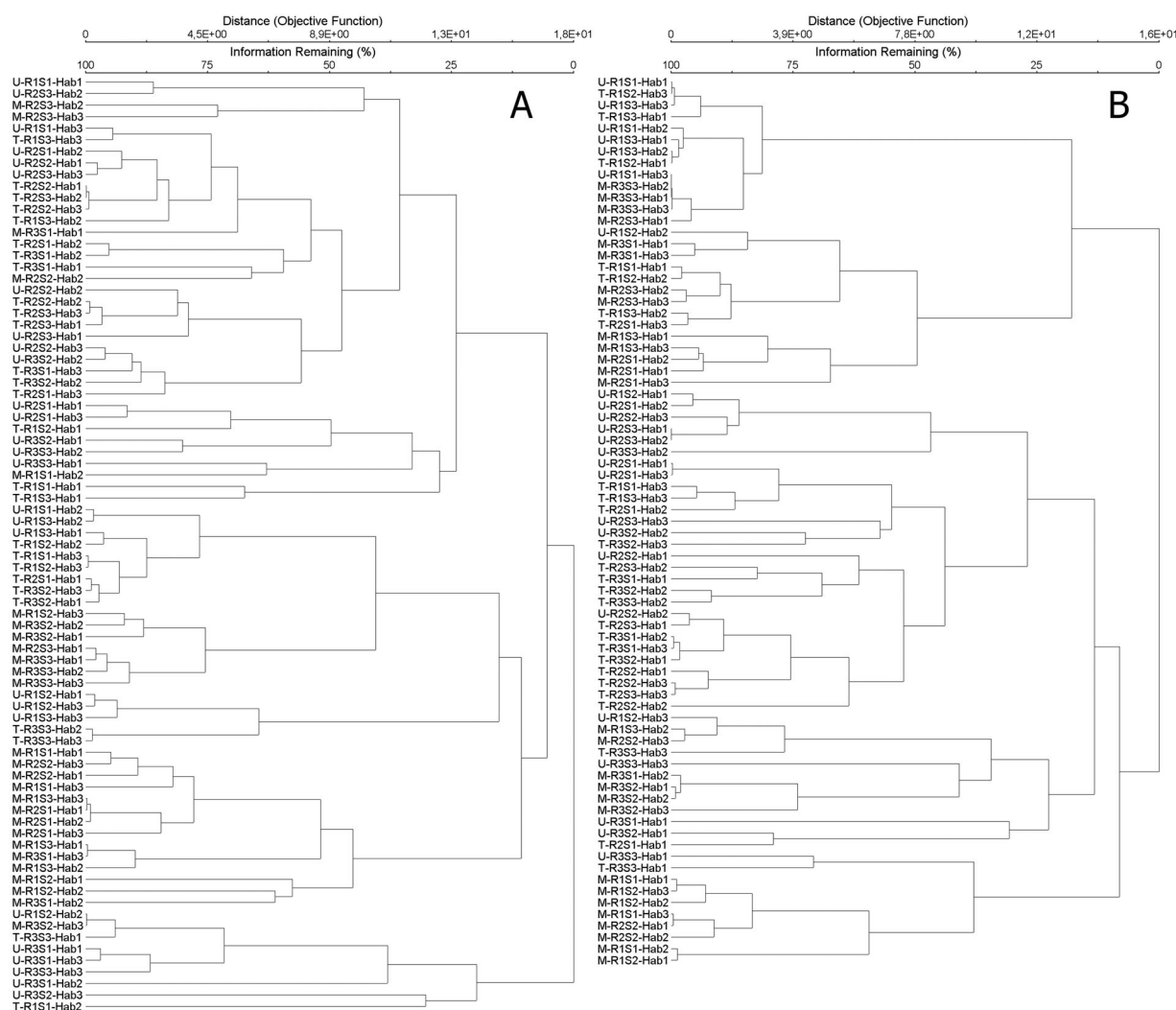


Figure 4. Hierarchical flexible beta clustering ($\beta = -0.25$) based on Bray-Curtis distance of (a) macroinvertebrates and (b) Chironomidae in the Paraíba River during the 2015 wet season. Tag codes indicate different sections of the river: U = upper segment, M = middle segment, and T = Taperoá River segment. R = reaches (1–3), S = sites (1–3), and Hab = habitat types (1–3). Percent chaining <4%.

alkaline waters (Santos and Eskinazi-Sant’Anna 2010), such as the study river catchment.

Hierarchical clustering and the MRPP showed that most spatial segregation in both macroinvertebrates and Chironomidae occurred at broader spatial scales (river segment and river reach), with finer scales less dissimilar. At broader scales, a range of environmental variables showed a great degree of variability associated with river hierarchy, such as altitude and river width, but also for some of the more local variables such as conductivity and the habitat structure. Such variability in the physical habitat available for colonization is frequently reported as an important factor establishing patterns of macroinvertebrate diversity and composition in streams (Heino et al. 2003, Clarke et al. 2008). Furthermore, the geographical distances between sites, associated with the patterns of temporal and spatial disconnectivity typical of intermittent rivers and streams, may also have caused

greater dissimilarities at the broader scales (Malmqvist 2002, Clarke et al. 2008, Rouquette et al. 2013).

Diversity partitioning provided further support for the importance of segments (β_4) and reaches (β_3) to the regional diversity pool because the observed β was higher than expected at these broader scales, and also explained a greater proportion of the total diversity for both macroinvertebrate and Chironomidae. Similar results were observed for zooplankton by Melo and Medeiros (2013) at the same river, where broader scales (in their case the pool/habitat scale) attributed greater contributions to the regional γ diversity. Those authors proposed that periods of low or no flow enhance diversity by segregating the plankton fauna composition among pools, therefore increasing β diversity at the pool/habitat scale.

A greater β component for macroinvertebrates at the broader scales has also been observed in temperate and

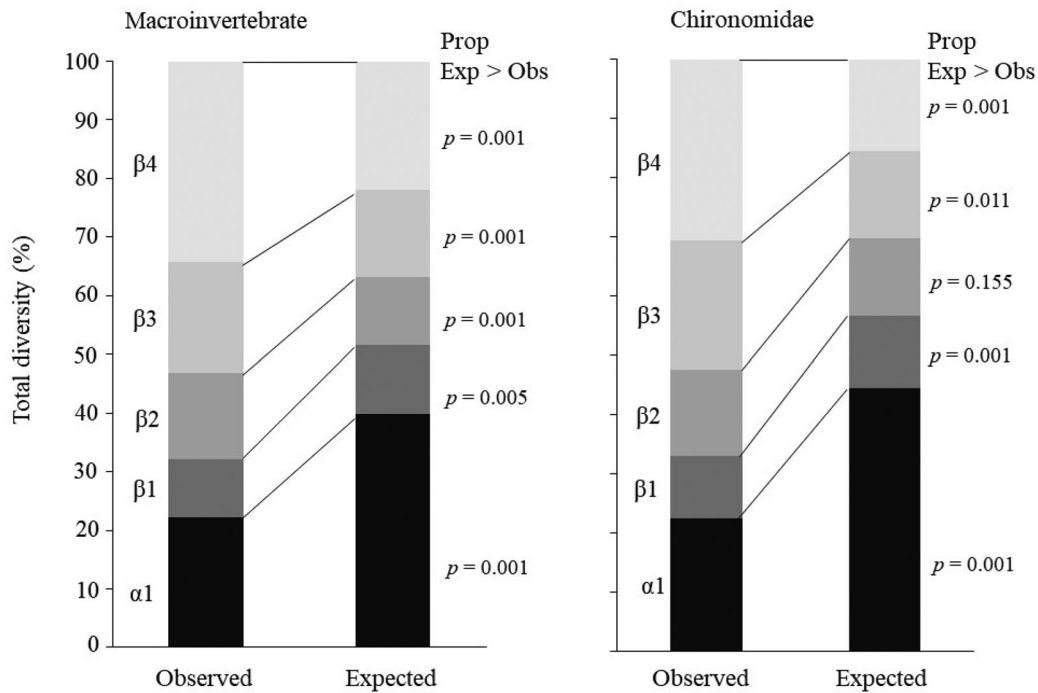


Figure 5. Observed and expected diversity for macroinvertebrates and Chironomidae, partitioned into 1 alpha and 4 beta components, expressed as percent of total species richness. p values indicate the proportion of randomized samples containing more species than the observed sample for each partition. $\alpha 1$: microhabitat species richness; $\beta 1$: beta diversity among microhabitats; $\beta 2$: beta diversity among pool/riffles; $\beta 3$: beta diversity among river reaches; and $\beta 4$: beta diversity among river sections.

tropical perennial streams (Heino et al. 2004, Robson et al. 2005, Ligeiro et al. 2010), where the finer scales added little heterogeneity to the species diversity. In dryland rivers, the temporal disconnectivity among patches of habitat has been associated with a decrease in local diversity, given the habitat reduction (Medeiros and Arthington 2014), dominance of generalist species (Sheldon et al. 2003), and distance from sources of organisms for recolonization (Vanschoenwinkel et al. 2009, Larned et al. 2010, Tonkin et al. 2016). As the time of disconnection among river reaches increases and pool size decreases, species less resilient or resistant to the hydrological disturbance would be progressively filtered from the regional pool (McKinney and Lockwood 1999, Chase 2007, Jacobsen and Dangles 2012). In such cases, a greater heterogeneity in species composition in broader scales, rather than between habitats within finer scales, should be expected. Other studies in temporary aquatic systems have shown that the variation in species composition and diversity can be higher at the finer scales, among pools ($\beta 2$), and/or among habitats within pools ($\beta 1$) (Sheldon et al. 2010). This phenomenon has been observed for macroinvertebrates in subtropical shallow lakes (Ávila et al. 2011) and zooplankton in dryland rivers (Melo and Medeiros 2013), contrary to the results observed in the present study.

Nonetheless, species may be replaced by other previously less abundant species in a successional pattern of filling available niches and therefore increasing diversity of species, as would be expected in environments mediated by hydrological disturbances where habitat heterogeneity leads to higher diversity. Studies have shown that the diversity of species of macroinvertebrates and Chironomidae (Rocha et al. 2012), zooplankton (Simões et al. 2008), and to a lesser degree fish (Medeiros and Maltchik 2001), actually increases during dry periods or during the pool phase. In the present study, because the sampling was performed during the rainy season, some degree of spatial connectivity likely homogenized species composition and diversity at the finer scales, therefore increasing the importance of greater scales to the diversity partitioning.

In conclusion, a combination of homogenization of assemblage composition associated with a reduced diversity at the finer scales may explain the greater contribution of broader scales to the diversity. This finding highlights the importance of the hydrological variability to the diversity of species in intermittent rivers, which can create conditions that select species from a larger diversity pool and break it into more specific assemblages at local scales, depending on the dry/wet seasonal dynamics (Usseglio-Polatera 1994, Medeiros et al. 2011, Melo and Medeiros 2013).

Disclosure statement

No potential conflict of interest was reported by the author(s).

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